

Quality contour

Gates and a doctor for AI coding agents

A rulebook and a set of gates that deploy into any repository and stop an AI agent from reporting green on something nobody checked – with a doctor that audits the gates themselves.

22

gates under a hard cap

22,600+

lines of canon

25

field findings, classified

1,113

behaviour tests on the canon

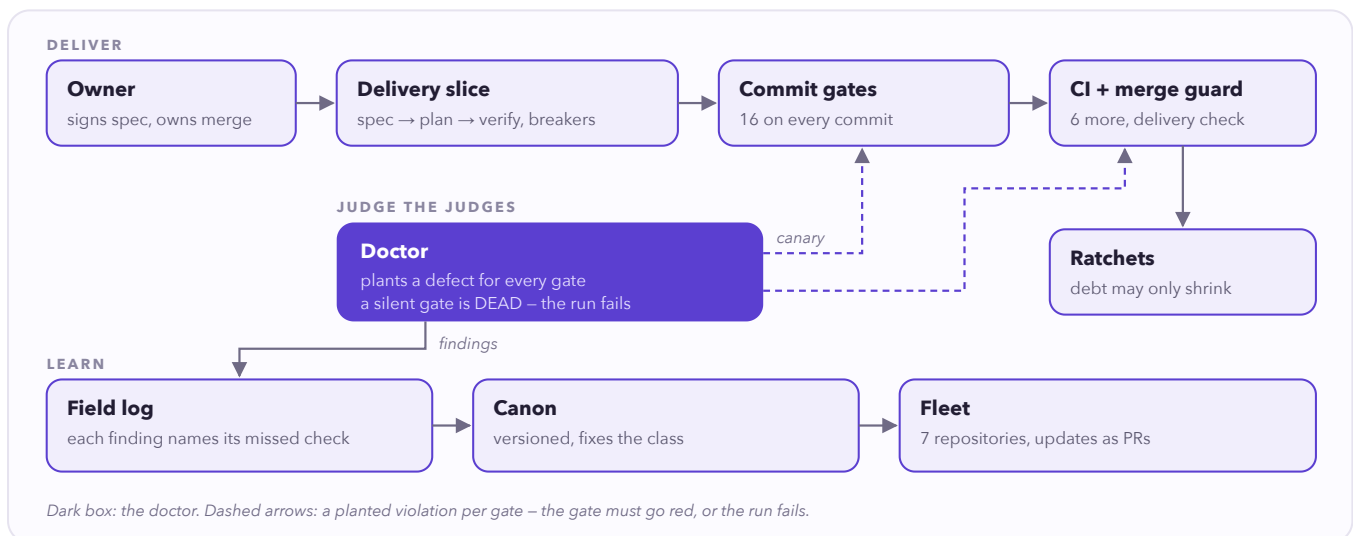
THE PROBLEM

An agent ships work that looks finished. The usual failure is not bad code – it is a green light over a check that verified nothing: a gate that exits 0 when its tool is missing, a mask copied from a template that doesn't know the repository's layout, a baseline that never shrinks. In one field test a suite with all 49 assertions deleted still passed 25 hooks at 93.91% coverage.

WHAT IT DOES

- Cuts work into size-capped slices – spec, plan, verify – with breakers that stop the agent
- Runs 22 gates, 16 on every commit and 6 in CI, under a hard cap
- Freezes existing debt in ratchets that may only shrink
- Audits every gate with a doctor that plants a defect and expects red

ARCHITECTURE



HOW IT FLOWS

- 1 The owner signs a spec; the agent builds inside a slice with size breakers.
- 2 Every commit runs the commit gates; growth over a ratchet snapshot is red.
- 3 CI runs the rest and the delivery check; main merges only when green.
- 4 The doctor plants a violation for each gate – a gate that stays silent is DEAD.
- 5 Any finding names the existing check that should have caught it.
- 6 The canon fixes the class and ships to every repository, verdicts before and after.

WORTH KNOWING

One digit makes a gate lie.

Raise a limit from 500 to 50000 and the hook still prints Passed; the doctor fails the run with DEAD.

Two gates judged wrong for a month.

Both were rated healthy. Known-answer tests now replay the bug and go red the day an update brings it back.

It publishes what it doesn't enforce.

Of 23 rules, 16 are held by a gate, 1 by a tool and 6 by review only – and the split is public.

DECISIONS THAT SHAPE IT

A hard cap on gates, not a gate per incident 01

Because process systems die of weight. A new gate needs a swap or a written justification, and the last two were each paid for by a measured defect class.

Cost: known defect classes can wait for a slot.

Ratchets instead of bans 02

Because legacy repositories must join on day one. Old debt is frozen in a snapshot that may only shrink; an upward re-snapshot fails.

Cost: it needs a remote baseline: pre-push or CI only.

A canary doctor, not a config reader 03

Because “wired in” is text; “actually judges” needs execution. Five dead gates surfaced in its first session.

Cost: it catches silence, not wrong verdicts – those need known-answer tests.

A decision log a checker can reject 04

Because a choice without its alternative can’t be audited. Every entry must read “chose X instead of Y – because Z”, enforced in CI.

Cost: it checks the form, not the truth.

Fix the class, not the instance 05

Because a fix in one gate left three neighbours with the same silence. Every finding now names the check that should have caught it, and a test enforces this.

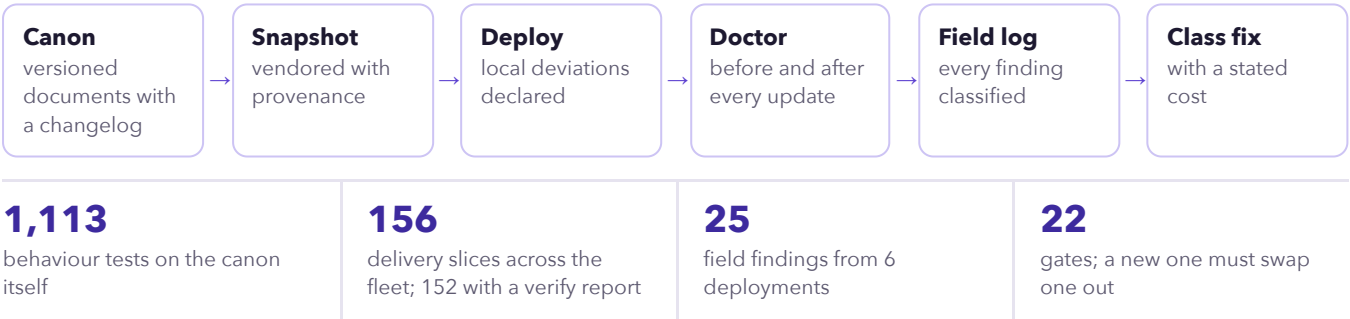
Cost: classes are named by hand.

Publish what is enforced and what is advisory 06

Because an unwired gate once let acceptance show all green. A meta-gate now fails any unwired script, and the enforced-versus-advisory split is public.

Cost: the meta-gate sees wiring, not code.

HOW IT IS ENGINEERED



Verdicts are named, not implied. AUTO – the canary went red; WEAK – the gate runs but doesn’t judge, and says why; ABSENT, SKIP and TOOL are declared. DEAD – declared, wired and silent on its own canary – is the only lie and the only verdict that fails the run. The contour calls itself unfinished: its gate-defect share is 28% against a 20% target.

WHERE IT TRANSFERS

Adoption is vendoring, not a framework install: a pinned snapshot of the canon, with provenance, is copied into the repository, and every local deviation is declared with a reason. Updates arrive as a pull request with doctor verdicts before and after, and a new revision reaches one canary project first. It scales across teams because its cost is capped by a gate budget, legacy code joins on day one through ratchets, and the doctor lets people who didn’t write a gate audit it.

STACK

Core Python, Bash, pre-commit, GitHub Actions · **Python gates** ruff, mypy --strict, import-linter, pytest-cov, mutmut · **TypeScript gates** ESLint, Prettier, dependency-cruiser, Stryker · **Both** jscpd, detect-secrets